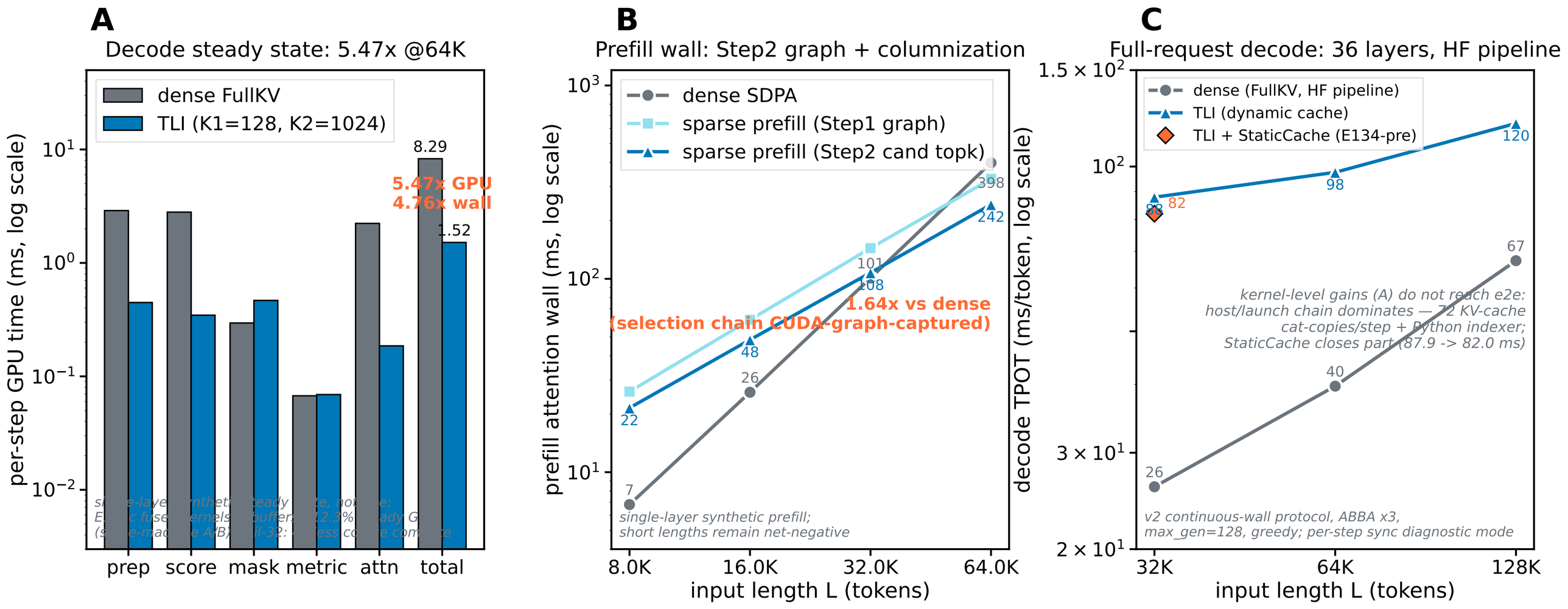
A
Decode steady state: 5.47x @64K
dense FullKV
TLI (K1=128, K2=1024)
per-step GPU time (ms, log scale)
8.29
1.52
5.47x GPU
4.76x wall
prep
score
mask
metric
attn
total
B
Prefill wall: Step2 graph + columnization
dense SDPA
sparse prefill (Step1 graph)
sparse prefill (Step2 cand topk)
prefill attention wall (ms, log scale)
decode TPOT (ms/token, log scale)
1.64x vs dense
(selection chain CUDA-graph-captured)
single-layer synthetic prefill;
short lengths remain net-negative
8.0K
16.0K
32.0K
64.0K
input length L (tokens)
C
Full-request decode: 36 layers, HF pipeline
dense (FullKV, HF pipeline)
TLI (dynamic cache)
TLI + StaticCache (E134-pre)
kernel-level gains (A) do not reach e2e:
host/launch chain dominates — 72 KV-cache
cat-copies/step + Python indexer;
StaticCache closes part (87.9 -> 82.0 ms)
v2 continuous-wall protocol, ABBA x3,
max_gen=128, greedy; per-step sync diagnostic mode
32K
64K
128K
input length L (tokens)